

U.S Citizen

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EDUCATION

Northwestern University

Expected June 2027

Bachelor of Science in Computer Science | Major in Economics

GPA: 3.88/4.0

- **Relevant Coursework** - Data Structures and Algorithms, Machine Learning, Computer Systems, Discrete Math, Computer Networking, Data and Information Processing, Design and Analysis of Algorithms

EXPERIENCE

Lab

Apr 2025 - Present

Undergraduate Researcher

Evanston, IL

- Constructed neural network architecture for pollinator detection with OpenCV and Python
- Collaborated with researchers at Chicago Botanic Garden to create a Python pipeline using Raspberry Pi and arducams connected through GPIO pins to enable researchers to log video captures with a button
- Established CI/CD pipeline changes and wrote unit tests, functional tests, and integration tests for deployment

May 2024 - Sep 2024

Software Engineering Intern

Milwaukee, WI

- Hosted weekly tutoring sessions to educate children on topics such as python and math
- Built interactive website in React deployed on AWS EC2 growing web traffic from 200 to 1000 monthly visitors
- Spearheaded the design and implementation of REST API endpoints using Express.js and JWT for secure authentication to manage mailing lists, volunteers, and student signups, reducing API response time by 33%
- Optimized AWS RDS database performance, reducing query times by 40% with indexing and normalization

PROJECTS

NBA Clip Finder

- Developed a full-stack application using React.js, Flask and MongoDB to return any play for an NBA player, enabling editors to find specific clips 200% faster than manually searching
- Reduced response times via multithreaded NBA API requests, honed MongoDB with caching techniques for database handling, increasing API lookup speeds by 50%

Stats GUI

- Built an interactive Vue.js application for 50 users to view player game data, achieving a 2X improvement in win loss ratio by detecting players to avoid while starting games
- Created automated minecraft accounts in JavaScript to scrape player packets and store them in MongoDB, capturing 10+ custom data points for 400,000 players
- Designed REST API in Node.js, Express.js and MongoDB handling 50,000+ daily GET and POST requests used Redis caching to reduce database read loads and response times

Python Compiler

- Created custom compiler in C++ for Python with memory management and dynamic allocation
- Implemented tokenizer, scanner, parser, and interpreter to handle and execute structured programs supporting loops, booleans, variables, conditionals, python pointers, and inputs through recursive-descent parsing
- Developed a command line debugger, supporting steps, breakpoints, variable states, and memory dumps to view current program execution

SKILLS

Programming Languages: Python, C++, C, JavaScript, Java, TypeScript, Go

Libraries/Frameworks: React, Node, Express, Vue, MongoDB, OpenCV, Flask

Tools: Git, Jenkins, Jira, AWS, Linux, Docker, Firebase, SQL, NoSQL, Excel, REST API